

Coffee Quality Prediction: SVM & XGBoost Analysis

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2026-01-03

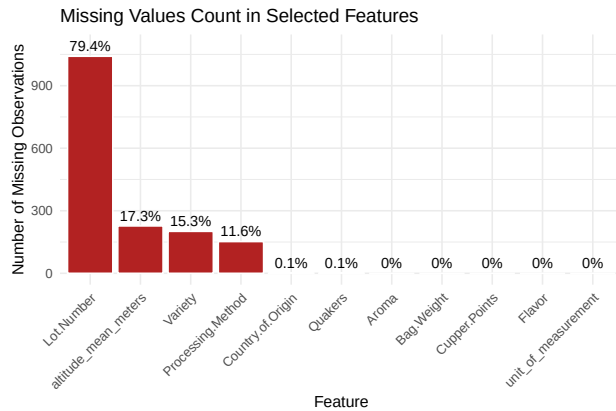
1 Introduction

In this report we set out to explore a data set about coffee quality. It covers everything from initial descriptive analysis to the mathematical foundations and application of two machine learning methods (SVM for Regression and XGBoost), as well as explainable AI analyses. We showcase how we use these techniques to extract meaningful insights into what makes a coffee expert choose their favorite coffees.

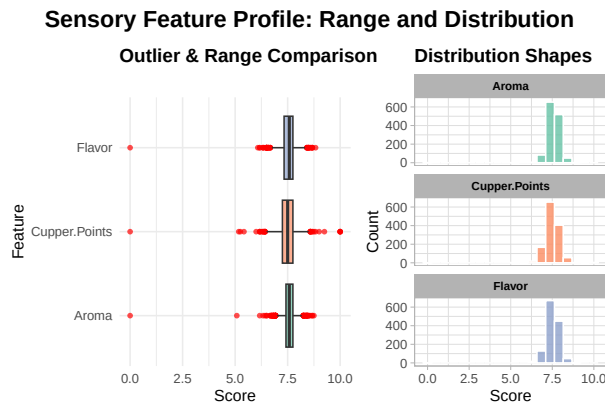
1.1 Descriptive Analysis

Summary of the Descriptive Analysis: The data used in this report originates from the website of the Coffee Quality Institute and was scraped in 2018 by user jldbc. It can be found on GitHub ¹.

The dataset has 1,311 coffee evaluations (observations) with 44 different variables. They features comprise a mix of sensory scores (e.g. Aroma, Acidity, Moisture) and agricultural cultivation data (e.g. Region, Altitude, Processing Method). Out of these, 24 variables are categorical and 20 are numerical. For our project, we decided to use **Cupper Points** as our target variable. “Cupper Points” is a score reflecting how much a professional coffee taster enjoyed an individual coffee. We avoided “Total Cup Points” as target variable as it is a linear combination of other sensory scores and hence not suitable for analysis.

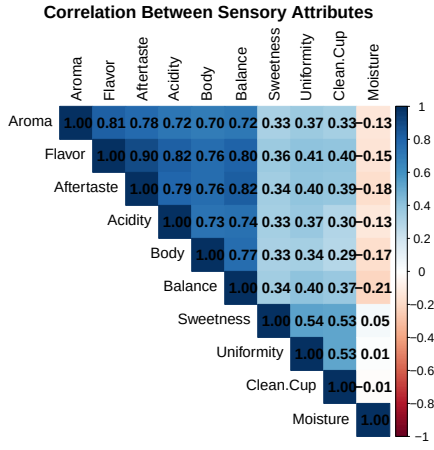


Dealing with Missing Data The amount of data missing from the data set varies greatly by feature. Especially for categorical data, we can see high ratios of up to almost 80% gaps. Variables with this high rates of missing entries will most likely not add useful signal to the model training. Instead we focus on features where either data is abundant or where only a medium amount (<20%) of data is missing and there are sensible ways to impute missing values.



The Distribution of the Variables Looking at the histograms for the sensory variables, we can see them following mutually similar distributions. A majority of the scores fall closely together in the interval between 7 and 8. A few outliers are noticeable in the extremes of the possible values. These could indicate a coffee taster having a strong opinion about the coffee, a special variety of coffee, or might be an error in the data.

¹<https://github.com/jldbc/coffee-quality-database>

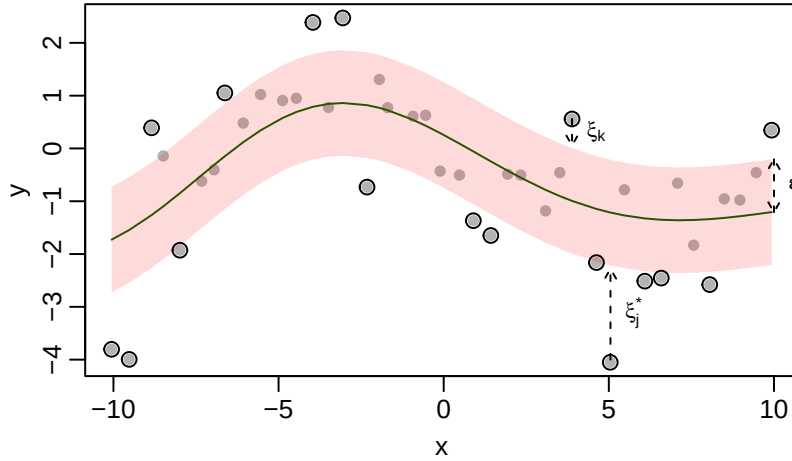


Correlation Between Numerical Features

Most of the sensory variables are highly correlated. For instance a pleasant smell, which translates to a high value for the Aroma score, is almost always accompanied by great taste as indicated by a high flavor score. Moisture stands out as the only sensory attribute that shows a slight negative or no correlation with all other variables in this group.

2 A Brief Mathematical Overview of SVM and XGBoost

2.1 Support Vector Machine for Regression



Support Vector Regression expands Support Vector Machines from discrete classification to a continuous regression model (Drucker et al. 1996).

For the regression task, instead of fitting a hyperplane, we are trying to fit a (multidimensional) function $f(x)$ to the data. Similar as in the classification task, we draw a margin around the function. However, now the margin ε is fixed at the start. (Note that ε here, is not the slack variable as the notation is commonly used for SVM in classification, e.g. in “An Introduction to Statistical Learning” (James et al. 2021a), but rather defines the width of the margin. Also, we do not aim to optimize the margin boundaries here, but instead fix them before computation starts). We want to find a function $f(x)$ that is as flat as possible, while fitting all points into the required margin. Again, we allow for some slack using slack variables. A wider margin decreases the risk of overfitting, while a smaller margin captures more intricacies of the data.

Mathematically speaking, want to find a function $f(x)$ with $|f(x_i) - y_i| \leq \varepsilon$ for all i . To avoid overfitting, we also introduce the constraint of making the function as flat as possible. “Flatness” in this context is a measure of how sensitive the function is to change. For a linear function $f(x) = x^\top \beta + b$ increasing this flatness can be expressed through minimizing the norm of the function gradient: $\|\beta\|$. Minimizing the expression $\frac{1}{2}\|\beta\|^2$ leads to the same optimum, while allowing for more elegant mathematical solutions, and is therefore used for computation.

Again, we want to allow for some slack, so as in the SVM for classification, we are introducing slack variables ξ_i and ξ_i^* . These slack variables allow for some room for margin violations. The optimization problem we want to solve is to minimize:

$$\frac{1}{2}\|\beta\|^2 + K \sum_{i=1}^n (\xi_i + \xi_i^*)$$

subject to:

$$y_i - f(x_i) \leq \varepsilon + \xi_i,$$

$$f(x_i) - y_i \leq \varepsilon + \xi_i^*$$

and

$$\xi_i > 0, \xi_i^* > 0 \quad \forall i$$

where K is fixed and β , ξ and ξ^* are variable. The factor K is introduced as cost for regularization purposes. It is chosen a priori to determine how strong the tradeoff between flatness and overfitting should be. A larger K leads to a stronger impact of the term, therefore penalizing stronger margin violations more heavily and resulting in a less flat function curve (higher bias, lower variance). Conversely, a smaller K leads to a flatter curve (lower bias, higher variance).

As in the SVM for classification, this can be generalized to non-linear functions by transforming the features using kernels.

2.2 XGBoost

XGBoost is an algorithm based on gradient boosted trees and second-order approximation (Chen 2016). It is widely used in machine learning due to its flexibility and strong empirical performance. These properties arise from a combination of algorithmic and systems-level optimizations, including efficient handling of sparse input data, approximate tree construction via weighted quantile sketches, and parallelized tree learning. In addition, careful optimization of memory access, data compression, and out-of-core computation enables the method to scale to very large datasets while maintaining computational efficiency.

This section describes the mathematical foundations of the algorithm, namely ensemble methods, regularization, and gradient boosting machines combined with second order approximation.

Tree Ensembles and Regularization

XGBoost models the prediction as an additive ensemble of K functions:

$$\hat{y}_i = \sum_{k=1}^K f_k(x_i), \quad f_k \in \mathcal{F},$$

where each f_k is a regression tree belonging to the function space $\mathcal{F}$ (Chen 2016; James et al. 2021b). Each tree acts as a weak learner that partitions the feature space into disjoint regions and assigns a constant prediction to each leaf (James et al. 2021b).

The model is trained by minimizing a regularized objective function of the form

$$\mathcal{L} = \sum_i \ell(y_i, \hat{y}_i) + \sum_{k=1}^K \Omega(f_k),$$

where $\ell(\cdot)$ is a differentiable loss function and $\Omega(f)$ is a regularization term that penalizes model complexity (Chen 2016). This term is defined as

$$\Omega(f) = \gamma T + \frac{1}{2} \lambda \sum_{j=1}^T w_j^2,$$

with T denoting the number of leaves in the tree and w_j the weight of leaf j . The regularization term encourages simpler trees and smooth leaf weights, helping to prevent overfitting.

The ensemble structure allows the model to reduce bias by combining multiple weak learners, while regularization controls variance by limiting tree complexity and the magnitude of leaf weights (James et al. 2021b). This balance between bias and variance enables XGBoost to achieve strong predictive performance while maintaining good generalization.

Gradient Boosting Machines

The tree ensemble is constructed in a stage-wise manner, where weak learners are added iteratively to improve the model (Friedman 2001). At iteration t , the prediction is given by

$$\hat{y}_i^{(t)} = \hat{y}_i^{(t-1)} + f_t(x_i),$$

with $f_t \in \mathcal{F}$ denoting the newly added regression tree.

Gradient tree boosting can be interpreted as gradient descent in function space, where each new learner is chosen to minimize the loss function in the direction of the negative gradient (Friedman 2001). Specifically, the new tree at iteration t is trained to fit the negative first order gradient

$$-g_i^{(t)} = - \left. \frac{\partial \ell(y_i, \hat{y})}{\partial \hat{y}} \right|_{\hat{y}=\hat{y}_i^{(t-1)}}.$$

XGBoost extends this framework by explicitly approximating the objective function using a second-order Taylor expansion of the loss around the current predictions:

$$\ell(y_i, \hat{y}_i^{(t-1)} + f_t(x_i)) \approx \ell(y_i, \hat{y}_i^{(t-1)}) + g_i f_t(x_i) + \frac{1}{2} h_i f_t(x_i)^2,$$

where

$$h_i = \left. \frac{\partial^2 \ell(y_i, \hat{y})}{\partial \hat{y}^2} \right|_{\hat{y}=\hat{y}_i^{(t-1)}}$$

is the second order derivative of the loss (Chen 2016). By incorporating both first- and second-order information, XGBoost enables efficient optimization of tree structures and leaf weights within each boosting iteration.

3 Support Vector Regression

3.1 Data Preprocessing and Cleaning

3.1.1 Feature selection

In SVM Regression, categorical variables are One-Hot Encoded using dummy variables. This means, features with many unique values will lead to a sparse feature matrix. This is disadvantageous as it increases computational cost and poses the risk of overfitting. Additionally, categories with only a few or even only one observation most likely will not add to model robustness, but are adding noise to the signal. Therefore, we are removing categorical features with many unique values, only keeping columns with a low number of unique values. E.g. we are not including the “Region” column with 344 unique values in the model training, because we’d on average have less than 4 observations per Region. For features with a high number of missing values, we either remove the column or impute the missing values. For instance for the “Variety” column, we have about 15% missing values and 8% entries “other” at 30 different varieties. Keeping those, would mean adding 30 sparse columns, and at the same time the information gain is most likely not too high with this many missing values. Therefore, we are not including this feature.

3.1.2 Imputation

For the altitude column, we have about 17% missing values. Here, we are imputing the missing values from the “Region” column. For each observation with missing altitude, we are imputing the median altitude of all coffees from the same region. This covers almost all missing values. For the remaining observations, we are imputing the mean altitude of all coffees from the same country of origin. As the “Region” and “Country” columns are not used in the model training, we are not creating correlated features with this imputation. We are imputing the missing values after the train/test split, using only values from the training set, to avoid data leakage from the test set into the training set.

3.1.3 Further preprocessing

The bag weight column contains both pounds and kilograms, so we are converting all to kilograms. However, for some entries, it is unclear which unit is used. For these, we are averaging the weight in pounds and kilograms. This means for those entries the weight is certainly incorrect, but should be within an order of magnitude of the correct weight. If we can assume that specialty coffee is sold in smaller quantities, we can still gain valuable information from the weight.

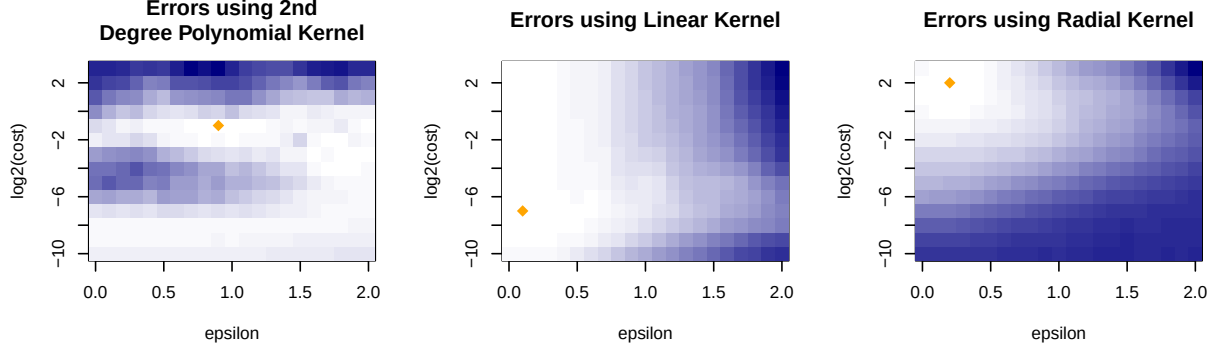
3.1.4 Scaling

The SVM method from the `e1071` package used for model training has a built in scaling functionality. Both predictor and target variables are scaled to have expected value 0 and variance 1. (Meyer et al. 2025) This transformation is based on the training data and applied to train and test data sets.

3.2 Hyperparameter Optimization

For the prediction of cupper points we train an SVM for regression on the data set, holding out the test data set for final evaluation. To choose appropriate hyperparameters, we conduct a hyperparameter optimization on the

training data set using 10-fold cross-validation. We use a grid search for different values for cost and epsilon for a radial, linear, and 2-degree polynomial kernel. We compare the mean absolute error between runs to choose the best-performing hyperparameter combination. Finally, we train a model with these hyperparameters on the entire training set and evaluate on the test data set.



These figures show the mean absolute error for the different hyperparameters by kernel. The best performing hyperparameter combination for each kernel are marked with orange points.

epsilon	cost	kernel	CV MAE
0.1	0.0078125	linear	0.208

The best performing kernel is the linear kernel, which means the relationship between the features and the target variable is best approximated by a multidimensional hyperplane.

4 XGBoost Model

4.1 Data Preprocessing

To ensure that the SVM and XGBoost models are comparable, we apply the same data preprocessing described above to XGBoost, except we skip feature scaling.

SVM requires scaling because it calculates distances between data points to find the optimal separating hyperplane. Features with larger ranges would dominate the distance calculations, which would make the model incorrect. XGBoost uses decision trees that select split points by maximizing information gain. Because these splits depend only on whether values are higher or lower than a threshold (e.g., “is Flavor > 7.5?”), feature scaling is unnecessary.

4.2 Configuration & Data

Config	Value
Seed	123
Split	80/20
CV Folds	10
Early Stop	50
Max Rounds	2000

Data	Value
Train n	1047
Test n	262
Features	21
Target $\bar{x}$	7.5

Model	MAE	R^2
Mean	0.309	0
Linear	0.147	0.59

Baseline models

We use two baseline models: the Mean Predictor and Linear Regression.

The **Mean Predictor** has the $R^2 = 0$, which confirms that our target has meaningful variance to explain. The MAE of 0.309 means that without any features, we’d be off by about 0.3 points on average.

Linear Regression provides an interpretable baseline with R^2 of 0.59, showing that 59% of the variance in cupper points can be explained by linear combinations of features. Any improvement from the XGBoost models means capturing non-linear relationships that linear regression misses.

4.3 Hyperparameter Optimization

eta	max_depth	subsample	colsample	rounds	CV MAE
0.01	5	0.7	0.7	360	0.156

Hyperparameter selection

We tune four main parameters to balance bias and variance:

eta (learning rate): 0.01–0.1. Smaller values make each tree contribute less, reducing overfitting but needing more rounds.

max_depth: 3–7. Controls tree complexity. Shallow trees capture simple patterns; deeper trees can model complex patterns but may overfit, especially on our dataset with about 1000 samples.

subsample: 0.7–0.9. Fraction of rows used per tree. Reduces overfitting by adding randomness. We avoid below 0.7, as it would leave too few samples per tree.

colsample_bytree: 0.7–0.9. Fraction of features per tree. Prevents any one feature from dominating.

These ranges are conservative. With about 1000 samples, extreme values (e.g., eta=0.3, max_depth=10) would overfit, but mild regularization is enough.

Lambda and Min_Child_Weight Regularization

We tested additional hyperparameters, but they were ineffective. XGBoost’s lambda (L2) and min_child_weight provided no benefit because implicit regularization from learning rate, max_depth, subsample, and early stopping already prevents overfitting. Extra regularization slightly worsened the metrics, so it was omitted.

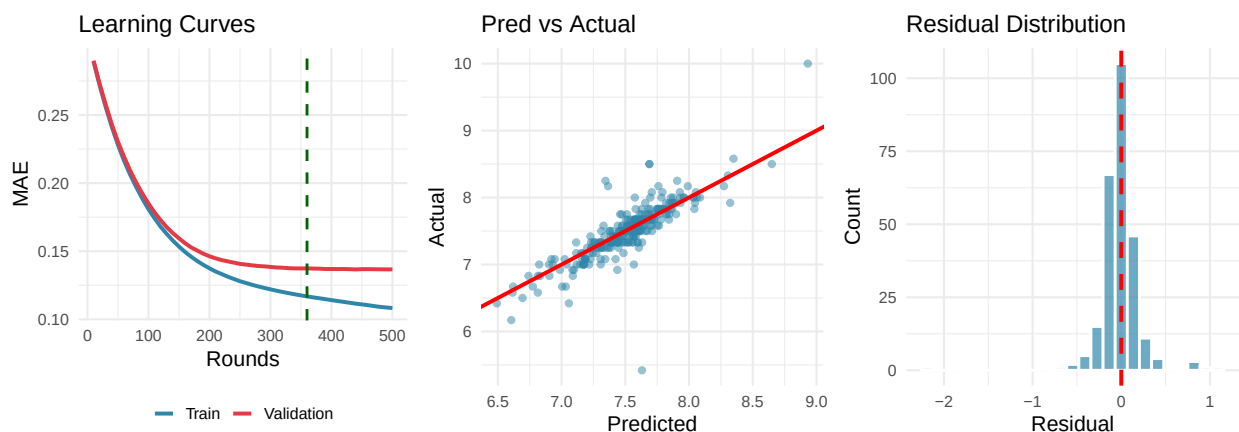
Bayesian Optimization vs Grid Search

We also tested Bayesian Optimization, but it was ineffective. With only 4 parameters and a small search space, it added overhead, slowed hyperparameter tuning, and offered no meaningful improvement. Grid search is sufficient for this dataset and model.

10-fold CV: Matches SVM for fair comparison. Each fold has about 100 samples, giving stable MAE estimates.

Early stopping (50 rounds): Stops training if validation MAE doesn’t improve for 50 rounds, automatically choosing the optimal number of trees and avoiding overfitting.

4.4 Diagnostic Plots



The **learning curves** show that the model converges with minimal overfitting. In the first 50 rounds, training and validation MAE drop as the model captures the main signal. After that, improvement slows and additional rounds provide little benefit, which justifies early stopping. The small train-validation gap of 0.021 MAE indicates that overfitting is minimal.

The **predictions vs actual** plot shows that most points lie close to the diagonal, indicating accurate predictions. Larger errors appear at the highest and lowest quality scores, which is expected because extreme values are rare and the model has fewer examples to learn from. There is no visible systematic bias, as points are evenly distributed around the diagonal.

The **residuals** are centered around zero, confirming that the prediction bias is minimal. Their roughly bell-shaped distribution suggests that errors are mostly random rather than systematic, meaning the model has captured the main signal.

4.5 Conclusions

Model performance: XGBoost achieves an MAE of 0.137, meaning predictions are usually within 0.14 points of the true cupper score on a 10-point scale. This is a small but consistent improvement over linear regression

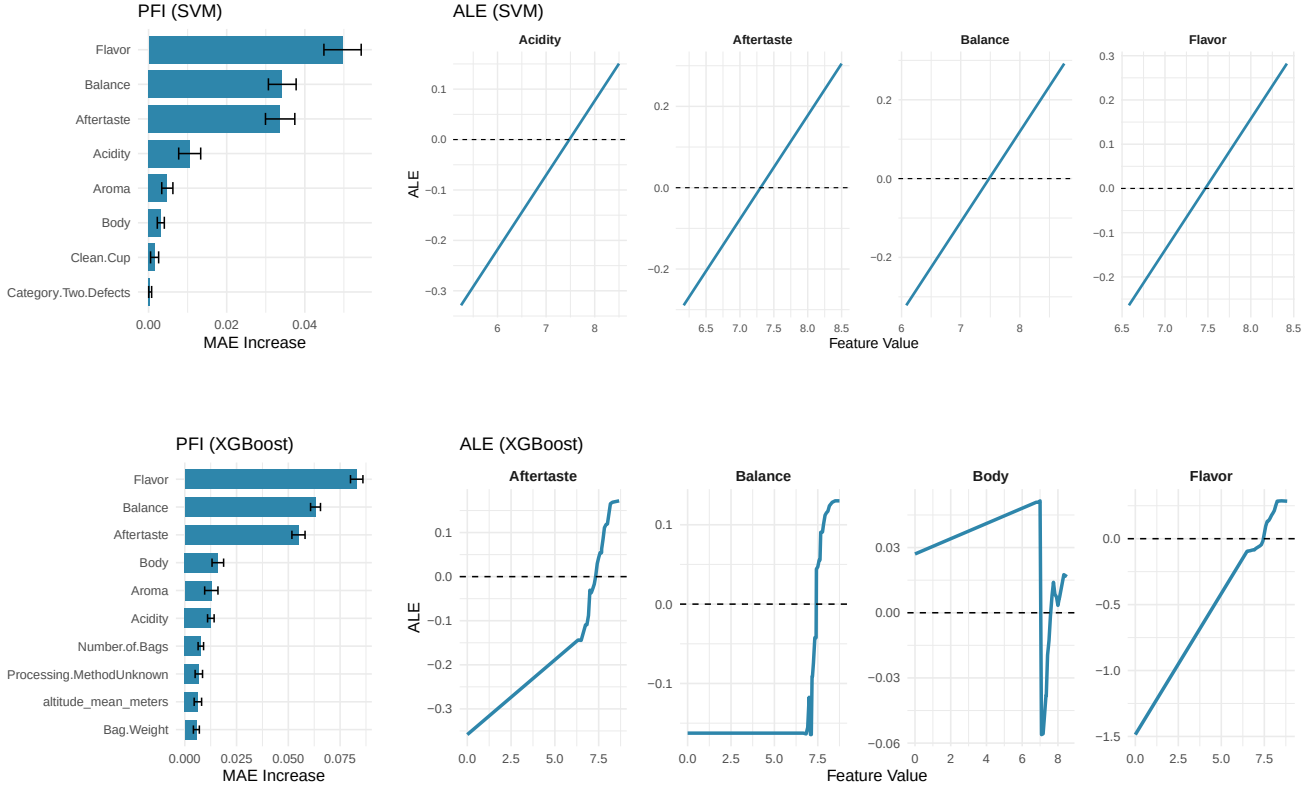
(MAE 0.147). The R^2 increases from 0.59 to 0.7, showing that XGBoost explains about 11% more variance.

Limitations: Performance is weaker at extreme quality scores due to limited data. The dataset size (about 1300 samples) may restrict learning of complex patterns, and temporal effects such as harvest year are not modeled (because of the issues with source data). Finally, the target variable cupper points is subjective and includes human rating variability.

5 Explainable AI

5.0.1 Permutation Feature Importance (PFI)

PFI measures how much model performance degrades when a feature’s signal is destroyed by shuffling (Friedman 2001). Unlike XGBoost’s native Gain metric, PFI is model-agnostic and accounts for feature interactions.



The top features are all sensory scores, confirming the model relies on expert taste evaluation.

5.0.2 Accumulated Local Effects (ALE)

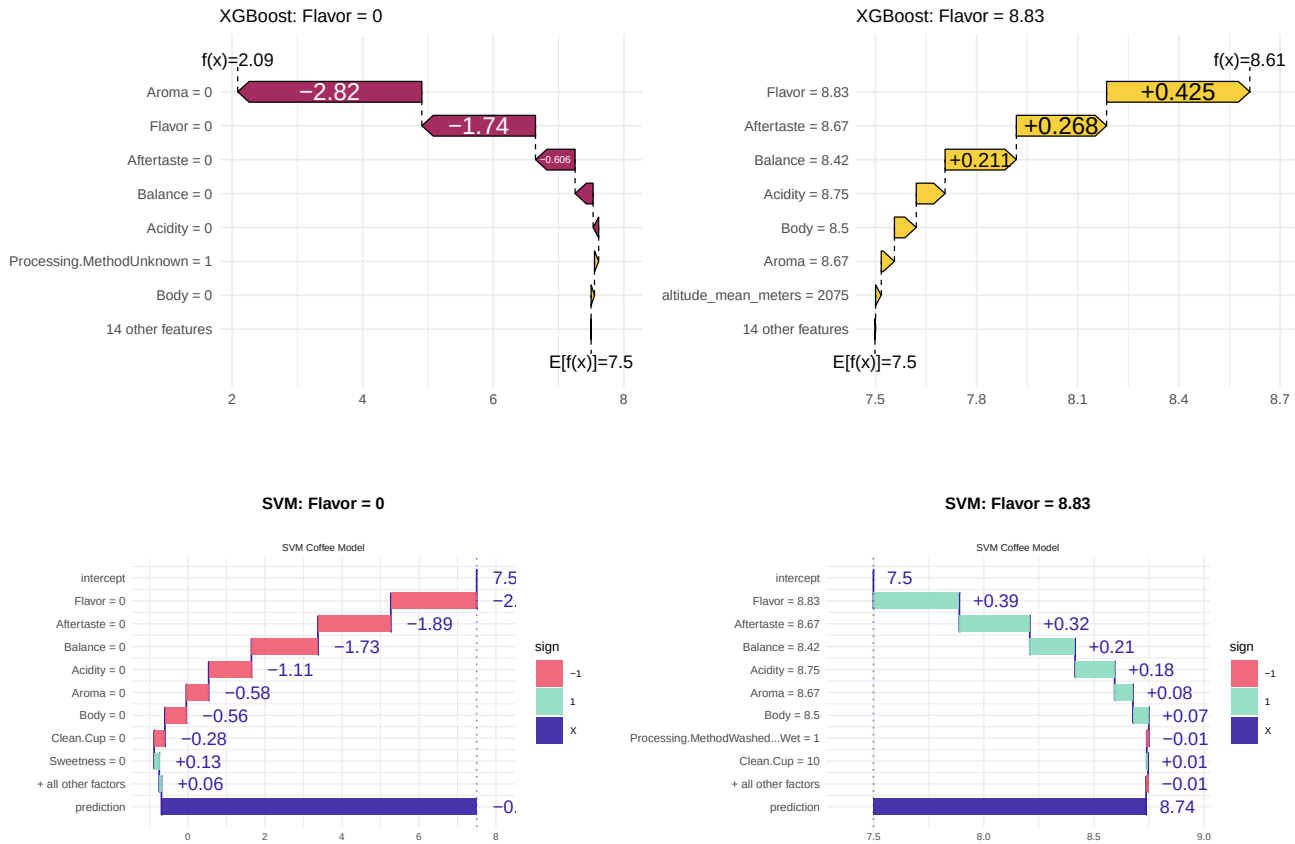
ALE plots show the marginal effect of a feature on the model’s predictions by averaging local changes in the prediction over the distribution of the other features (Apley and Zhu 2020). It is chosen here instead of PDP (Partial Dependence plots), because it is more robust to multicollinearities and centered around 0. We plot the top four features ranked by permutation feature importance (PFI).

All sensory features show the expected pattern that higher scores lead to higher predicted quality, only in Balance there is a sudden dip to a negative impact on the cupper score around values of 7, after which the effect goes back to being positive. The ALE range for Flavor is largest, indicating it has the strongest marginal effect on predictions.“

5.0.3 Shapley Additive Explanations (SHAP)

SHAP values provide a unified measure of feature importance based on cooperative game theory (Lundberg and Lee 2017). For each prediction, SHAP assigns each feature a value that represents its contribution to moving the prediction away from the baseline (average prediction). We utilize the SHAP values to inspect the microscopic

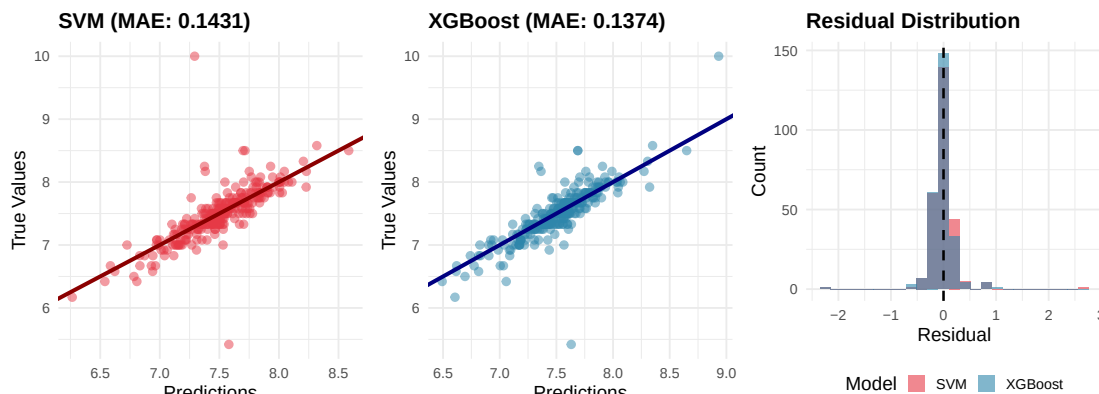
effects of the feature values for two specific observations. For this we chose one observation with a high value for Flavor and one with a low value, since this is the most important feature according to PFI.



The microscopic plots for both models show the expected effects of the Flavor feature on the overall prediction. For a Flavor of 0, both models strongly decrease their prediction, XGBoost by 1.74 and SVM by 2 cupper points. Interestingly, the XGBoost model derives an even greater effect (-2.82) from Aroma of 0 in this particular observation, even though Aroma is only the 5th most important feature according to PFI. The highest value of Flavor (8.83) leads to an increase of 0.425 cupper points in the XGBoost model and 0.39 in SVM. Since the sensory features are also highly correlated, their values are all relatively low or high in accordance to the Flavor feature. This leads to the effects being spread across the different features and could explain why in the first observation the aroma has an even stronger effect than the Flavor in the XGBoost model.

6 Model Comparison - SVM vs XGBoost

Model	MAE	RMSE	R2
SVM (Linear Kernel)	0.1431	0.2790	0.5855
XGBoost	0.1374	0.2385	0.6973



Both models achieve similar predictive accuracy, with XGBoost having slightly smaller MAE (0.137 vs 0.143) and larger R^2 (0.7 vs 0.59). The scatter plots confirm that both models produce predictions closely aligned with true values, with no systematic bias evident in either case. The residual distributions are nearly identical and centered around zero, indicating that both approaches effectively capture the underlying signal in the data.

6.1 Literature

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6.2 Statement of AI Use

Generative AI was used in this report to generate the R code for the analyses. Across all programming sections LLMs have been used to aid in code generation. The authors state that the generated code has been manually adjusted, verified and tested to ensure high quality. AI was also used to help formatting the report: writing LaTeX mathematical formulas and adjusting the tables size and formatting.

However, the text part of the report, the analysis of the plots, the research, the development the workflow etc. was done by hand using the stated sources as well as lecture material and examples from the exercises.